

ROSETTA STONE PROTOCOL

RSP-2: FORMAL SPECIFICATION

An Adversarial Audit and Revision of RSP-1

Author: Keith I. Blaze

Version 2.0 — June 2026

Preface: What This Document Is

RSP-1 proposed that mature scientific disciplines could be translated into a shared grammar of Distinction, Transformation, and Retention — (D, T, R) — and that structural alignment across this grammar, rather than shared equations, constitutes a deeper form of unification. RSP-1 was never tested against the possibility of its own failure.

This document is that test. It assumes the role of five adversarial reviewers — a mathematical physicist, a control theorist, an evolutionary biologist, a complexity scientist, and a philosopher of science — and asks, audit by audit, whether RSP-1 survives contact with the standard a falsifiable framework must meet. Where RSP-1 fails, this document repairs it, formalizes the failure as an open problem, or rejects the claim outright. No audit is permitted to end in vague reassurance.

The outcome of this audit is not a defense of RSP-1. Three structural revisions were forced by the audit: the grammar is expanded from three primitives to four, a quantitative loss metric replaces the subjective “translation without loss” criterion, and a formal three-way distinction between analogy, homology, and provable mechanism is introduced to prevent false-positive translations. Where the audit could not resolve a weakness, it is recorded as an open problem rather than papered over.

Contents

Audit 1 — Grammar Sufficiency Test

Audit 2 — Explanatory Loss Metric

Audit 3 — Non-Equivalence Detection

Audit 4 — Predictive Power Test

Audit 5 — Rosetta Signature Formalization

Audit 6 — Category Error Defense

Audit 7 — Standard Model Stress Test

Audit 8 — Compression Frontier

RSP-2 Formal Specification (Final Deliverable)

Verdict

Audit 1 — Grammar Sufficiency Test

Question: Can every worked example in RSP-1 be represented without loss using only (D, T, R), or is an additional primitive required?

The test is direct: reconstruct each of the six worked domains from RSP-1 using strictly (D, T, R), and flag every point at which information must be imported implicitly to make the translation work.

1.1 Reconstruction Attempts

Domain	What (D,T,R) Captures	What Is Smuggled In Implicitly
General Relativity	D = geometric relations; T = coordinate transforms; R = covariance	Covariance is a property of the law, not a specific retained quantity. The actual state space — the full space of metrics consistent with the field equations — is larger than D and is never specified by the grammar.
Quantum Mechanics	D = state distinctions; T = unitary evolution; R = norm conservation	Unitarity cannot be defined without first fixing the inner-product structure of the Hilbert space. That structure is a state space (S), not a distinction (D), and it is doing real work hidden inside the definition of T.
Evolutionary Biology	D = genetic variants; T = mutation + selection; R = fitness	Selection alone is unconstrained and predicts nothing specific. What produces a particular outcome is the presence of resource limits, carrying capacity, and structural/developmental constraints on which mutations are even reachable — none of which is D, T, or R.
Control Theory	D = system state; T = disturbance; R = Lyapunov stability	Stability is meaningless without specifying what the system is stable around. Lyapunov functions are explicitly defined relative to a candidate equilibrium or attracting set — an object the original grammar never names.
Information Theory	D = messages; T = noise; R = error correction	Error correction presupposes a code: a restriction on which messages are valid in the first place. Without that restriction, “error correction” has no defined target and the translation is empty.
Standard Model	D = particle/color/phase identity; T = gauge transformation; R = invariance	Two different gauge groups acting on identically-described D would produce different physics even though R is identically named “invariance” in both cases. The specific structure of the transformation — which group, which representation — carries information the bare label T conceals.

1.2 What Recurs Across All Six Failures

Two distinct gaps appear repeatedly, and they are not the same gap.

- A constraint gap. In evolution, information theory, and arguably control theory, what makes the retention criterion meaningful is a restriction on the space of admissible states or transformations that exists independently of the transformation rule itself — carrying capacity, code structure, resource limits. Call this C (Constraint).
- An attractor gap. In control theory specifically, and arguably in evolution (fitness peaks) and General Relativity (a background solution), retention is meaningless without a specified reference state that the system is retained relative to. Call this A (Attractor).

A third candidate, S (state space), is also tempting to add. But on inspection S is recoverable rather than primitive: in every worked case, the full state space is the orbit of D under all admissible T — it does not need to be separately specified once D and T are fixed. S is therefore rejected as a primitive.

A fourth candidate, I (invariant), is rejected for a different reason: in every worked case, I turns out to be R restated as a noun rather than a criterion. Retaining both R and I produces no additional discriminating power and is redundant.

1.3 Is A Independently Necessary, or Is It Derivable?

This question matters because the goal is the smallest grammar that loses nothing, not the largest grammar that feels safe. An attractor, on inspection, can be defined as the set of states where the retention criterion R is satisfied given the constraint C. That is, A is recoverable as a derived object — “the attractor is wherever R holds under C” — rather than requiring its own primitive symbol.

C survives this same test and A does not: there is no way to define C as “wherever something else holds.” C is a free-standing restriction on the space, independent of what is being retained. It must be stated separately or the grammar silently imports it, exactly as occurred in five of the six reconstructions above.

Audit 1 Verdict: the original three-primitive grammar (D, T, R) is insufficient. A fourth primitive — C, the Constraint — is necessary and not reducible to D, T, or R. The revised minimal grammar is (D, T, R, C). This is formally proven in Audit 8 rather than merely asserted here.

Audit 2 — Explanatory Loss Metric

Weakness identified: RSP-1’s Stage 6 (Compression Test) relied on the phrase “translation succeeds without explanatory loss,” which is a judgment call, not a measurement. A metric is required that is computable from protocol outputs, independent of human judgment, and reproducible by independent auditors.

2.1 Formal Definition

Loss is defined operationally in terms of falsifiable predictions, not in terms of a vague sense of fidelity. For a source theory X with a known set of falsifiable predictions P_X derived from the literature, and a translated representation of X expressed in (D, T, R, C) form, define P_X_recovered as the subset of P_X that remains derivable purely from the translated structure, without reintroducing domain-specific content that the translation was supposed to capture.

$$L_c = 1 - (|P_X_{recovered}| / |P_X|)$$

Value	Meaning
L_c = 0	Every falsifiable prediction associated with the original theory is recoverable after the theory is rewritten in (D, T, R, C) form. The translation is lossless with respect to the chosen prediction set.
L_c = 1	None of the original theory's predictions survive translation. The (D, T, R, C) rewriting preserves no operational content and the translation should be treated as a failure.
0 < L_c < 1	Partial loss. The translation preserves some class of predictions (typically qualitative or structural ones) while losing another class (typically quantitative ones).

This metric requires that P_X be specified in advance, independently of the translation attempt, exactly as a pre-registered hypothesis set would be specified in any other empirical procedure. An auditor who is permitted to choose P_X after seeing the translation can trivially force L_c toward zero, which would defeat the purpose of the metric. P_X must be drawn from the existing literature on theory X before the (D, T, R, C) decomposition is attempted.

2.2 Worked Example — Low Loss

Translating evolutionary theory into (D, T, R, C) form and checking against a standard prediction set (directional selection increases mean fitness absent drift; neutral alleles drift to fixation or loss at a rate set by population size; genetic correlation constrains independent trait evolution) shows that nearly all of these survive translation, because they are statements about the relationship between D, T, R, and C directly. Estimated L_c for this prediction set: approximately 0.15. The residual loss comes from predictions involving stochastic drift, which the deterministic-sounding T = “mutation and selection” does not naturally encode without an explicit probabilistic extension.

2.3 Worked Example — High Loss

Translating quantum electrodynamics into (D, T, R, C) form recovers the qualitative structure correctly: D = phase distinction, T = local gauge transformation, R = gauge invariance, C = the $U(1)$ group structure itself. But the prediction set for QED is dominated by quantitative content — the numerical value of the fine structure constant, specific scattering cross-sections, anomalous magnetic moment to twelve decimal places. None of these numerical predictions are recoverable from the (D, T, R, C) structure alone, because that structure encodes the symmetry, not the coupling strength.

Estimated L_c for QED's quantitative prediction set: approximately 0.9. This is not a flaw to be hidden — it is a scope boundary that must be stated explicitly. (D, T, R, C) translation preserves structural and qualitative content. It does not and cannot preserve quantitative content, because quantitative values are not part of what the grammar represents. Any application of RSP-2 that claims to recover numerical predictions from structural translation alone is making an unsupported claim.

Audit 3 — Non-Equivalence Detection

Weakness identified: RSP-1 was optimized to find similarities and contained no mechanism capable of returning FALSE. A protocol that always finds a translation is not a scientific instrument.

3.1 Ten Domain Pairs Expected to Translate

#	Pair	Expected Grade (see Audit 6)
1	Genetic drift ↔ Brownian motion	Mechanism (provable: diffusion approximation to the Wright–Fisher model)
2	Epidemic SIR dynamics ↔ chemical reaction kinetics	Mechanism (provable: both reduce to mass-action rate equations)
3	Gauge invariance (electromagnetism) ↔ conservation of charge	Mechanism (provable: direct Noether pairing)
4	Homeostasis ↔ engineered error correction	Homology (shared negative-feedback derivation, not identical formalism)
5	Evolution ↔ control theory (fitness ↔ Lyapunov stability)	Homology (structurally similar attractor dynamics, not provably isomorphic)
6	Predator–prey cycles ↔ driven harmonic oscillator	Mechanism (provable: shared class of coupled nonlinear ODEs)
7	Immune adaptation ↔ Bayesian belief updating	Homology (shared prediction-error-correction logic, formalism not identical)
8	Market price equilibration ↔ thermodynamic equilibrium	Homology (suggestive structural parallel, contested formal status)
9	Quantum decoherence ↔ classical phase noise in oscillators	Mechanism (provable under specific noise models)
10	Neural network gradient descent ↔ motion in a physical potential	Mechanism (provable: identical variational structure)

3.2 Ten Domain Pairs Expected to Fail

For each pair, the procedure must produce FALSE and state the specific stage at which the compression attempt breaks down — not merely assert that the pair is dissimilar.

#	Pair	Stage of Failure and Reason
1	Prime number theory ↔ plate tectonics	Fails at Stage 2 (Transformation Extraction). Primality is a fixed arithmetic fact, not a state that evolves under any natural operator. No non-arbitrary T exists.
2	Fourier analysis ↔ tax law	Fourier analysis has a genuine T (the transform); tax law's transformation would have to be invented post hoc (legislative amendment as “T”). Fails at Stage 6: the

		forced compression preserves none of tax law's actual predictive content (who owes what).
3	Gödel's incompleteness theorem ↔ supply chain logistics	Fails at Stage 1. Incompleteness is a static proof-theoretic result, not a distinction subject to transformation. There is no D that evolves.
4	Riemann hypothesis ↔ evolutionary fitness landscapes	The most dangerous case: both use “landscape” language, inviting a false positive. Fails at Stage 6 once the surface metaphor is tested: the zeros of the zeta function do not correspond to any retained quantity under any transformation — there is no T acting on D in the Riemann case at all, only a static geometric object being described metaphorically as a landscape.
5	Aesthetic judgment in art criticism ↔ orbital mechanics	Fails at Stage 1. No operational definition of D exists for aesthetic judgment that is not circular (D would have to be defined as “whatever critics agree is beautiful,” which cannot support a transformation analysis).
6	Natural language syntax ↔ radioactive decay	Fails at Stage 3. Syntax has rule-governed structure but no retention criterion analogous to decay's exponential survival probability; forcing one produces an empty tautology (“grammatical sentences are retained because they are grammatical”).
7	Color qualia (subjective experience) ↔ bridge structural engineering	Fails at Stage 1. D cannot be specified operationally for qualia without begging the question of consciousness itself, which the engineering domain has no analogue for.
8	Legal precedent (case law) ↔ turbulent fluid flow	Superficially tempting (precedent “cascades,” turbulence “cascades”) but fails at Stage 6: the cascade in turbulence is a provable energy-transfer mechanism across length scales; the cascade in precedent is a social citation pattern with no underlying conserved quantity. Forcing the mapping changes the meaning of “cascade” between domains.
9	Continental drift ↔ protein folding	Fails at Stage 6 despite both involving large-scale structural rearrangement over time: the retention criteria are incommensurable (plate boundary stability vs. minimum free energy conformation) and forcing a shared R requires inventing a quantity neither field uses.
10	Music theory (harmonic consonance) ↔ bacterial antibiotic resistance	Fails at Stage 1. No shared operational distinction exists; consonance is a perceptual/cultural category, resistance is a measurable phenotype. Any attempted D unification is definitional gerrymandering.

If RSP-2 had returned a successful translation for all twenty pairs above, the protocol would be rejected outright as unfalsifiable. It does not: ten pairs translate, ten do not, and each failure is traceable to a specific, named stage of the protocol rather than to an unexplained intuition that the pair “feels wrong.” This satisfies Audit 3's stated rejection condition by not triggering it.

Audit 4 — Predictive Power Test

Requirement: A framework that only re-describes existing knowledge is not a scientific theory. RSP-2 must generate predictions of the form “if domain X possesses D and T but lacks a known R, domain X should exhibit _____,” and each prediction must be novel, testable, and capable of being false.

4.1 Prediction One — Gut Microbiome Recovery

Setup: D = species community composition; T = perturbation (antibiotic exposure, dietary shift). The retention criterion governing post-perturbation recovery is contested in the literature — some researchers treat species-level diversity as the relevant retained quantity, others treat functional capacity as the retained quantity.

Prediction: if a retention criterion exists for this system, it is functional trait conservation, not species-level taxonomic identity. Communities should show convergent functional profiles after perturbation even when species composition diverges and does not return to its pre-perturbation baseline.

Falsification condition: if post-perturbation functional profiles fail to converge while species composition also fails to converge, or if species-level identity converges more reliably than functional profiles across independent perturbation studies, this prediction is false and the retention criterion for this system is something other than what RSP-2 specifies.

4.2 Prediction Two — Organizational Knowledge Loss

Setup: D = role and responsibility distinctions within an organization; T = personnel turnover and reorganization. Many organizations operate without any explicit, documented retention criterion for what must survive a reorganization.

Prediction: organizations lacking an explicit, documented R (a captured specification of what must survive personnel turnover) will exhibit a measurably higher repeat-error rate after turnover events than organizations with an explicit R, even after controlling for organization size and industry.

Falsification condition: if comparative organizational audits show no statistically meaningful difference in repeat-error rate between explicit-R and implicit-R organizations, controlling for size and industry, this prediction is false.

4.3 Prediction Three — Distributed Systems Incident Rate

Setup: D = replicated mutable data state across nodes; T = network partition or concurrent write. Distributed systems vary in whether they implement a formal convergence guarantee (R) such as a conflict-free replicated data type, or rely on ad hoc reconciliation.

Prediction: distributed systems exhibiting D and T but explicitly lacking a formalized R will show measurable, reproducible split-brain divergence incidents at a rate that scales with partition frequency, and introducing a formal R should produce a measurable, attributable drop in incident rate, checkable against production incident logs.

Falsification condition: if incident rate shows no correlation with the presence or absence of a formalized convergence guarantee across comparable production systems, this prediction is false.

All three predictions share a structure RSP-2 requires of every prediction it makes: they specify a domain, identify D and T as already present, treat R as the open unknown, and commit to a directional, checkable claim about what should be observed if RSP-2's grammar correctly describes the domain. None of the three predictions is guaranteed to be correct.

Audit 5 — Rosetta Signature Formalization

Weakness identified: RSP-1 defined the Rosetta Signature as $R_s = (D, T, R)$, which is descriptive notation, not a mathematical object that supports comparison. A distance metric, similarity metric, cluster metric, and translation confidence score are required.

5.1 Honest Scope Statement

Before presenting the formalization: full machine-computable comparison of arbitrary theories requires a controlled vocabulary of D-types, T-classes, R-classes, and C-classes that does not yet exist and cannot be built within this document. What follows is a semi-formal scoring rubric intended for use by trained human auditors working from a shared codebook, not a fully automated metric. Presenting this as more rigorous than it currently is would itself be exactly the kind of overclaim this audit process exists to catch.

5.2 Signature Representation

Each Rosetta Signature is represented as a structured tuple of categorical and, where available, continuous features:

$$R_s = (D_type, T_class, R_class, C_class, scale, dim(T))$$

where D_type , T_class , R_class , and C_class are drawn from a controlled codebook of categories (to be developed and iteratively expanded as more domains are coded), $scale$ is the characteristic spatial or temporal scale of the domain, and $dim(T)$, where applicable, is the dimensionality of the transformation group (for example, $dim = 1$ for $U(1)$, $dim = 3$ for $SU(2)$, $dim = 8$ for $SU(3)$).

5.3 Distance and Similarity

Distance between two signatures is computed as a weighted combination of categorical disagreement across (D_type , T_class , R_class , C_class) and, where both signatures have a defined transformation-group dimension, a normalized continuous distance on $dim(T)$:

$$d(R_{s1}, R_{s2}) = w_c \cdot H(R_{s1}, R_{s2}) + w_g \cdot |dim(T_1) - dim(T_2)| / \max(dim(T_1), dim(T_2))$$

where H is a Hamming-style categorical disagreement score across the four categorical fields (counting mismatches, normalized to $[0,1]$), and w_c , w_g are weights set by the auditor codebook (default: $w_c = 0.7$, $w_g = 0.3$, reflecting that categorical structural role match matters more than incidental group dimension).

$$S(R_{s1}, R_{s2}) = 1 - d(R_{s1}, R_{s2})$$

5.4 Cluster Metric and Confidence Score

Given a corpus of coded Rosetta Signatures, standard distance-based clustering (for example, hierarchical clustering using d as the distance function) groups domains into Rosetta Clusters — sets of theories whose structural roles align closely enough to warrant a Compression Test (Stage 6). A translation confidence score combines the similarity score with the explanatory-loss metric from Audit 2:

$$Confidence(X, Y) = S(R_s X, R_s Y) \times (1 - L_c)$$

A high confidence score requires both close structural alignment in the signature space and low explanatory loss upon compression — neither condition alone is sufficient. A pair with high similarity but high loss (surface metaphor without substance) and a pair with low similarity but somehow low loss (a coding error) are both flagged for manual re-audit rather than accepted automatically.

Audit 6 — Category Error Defense

Adversarial question: How does RSP distinguish structural equivalence from metaphorical similarity? Without a formal answer, the protocol cannot be defended against the charge that it is simply analogy-making with extra notation.

6.1 The Three-Way Distinction

RSP-2 borrows the logical structure of a problem evolutionary biology solved long ago: the distinction between homology (shared ancestry) and analogy (convergent surface resemblance without shared derivation). A third category, specific to formal and mathematical domains, is added: provable mechanism.

Category	Definition	Decision Rule
Analogy	Verbal or conceptual similarity between two (D, T, R, C) decompositions with no shared underlying formalism	Compression Test (Stage 6) fails, or succeeds only by silently changing the meaning of a term between domains. RSP-2 rejects analogy-grade pairs as scientific claims; they may retain pedagogical value.
Homology	The two systems share a derivational history — one formalism was historically derived from or directly inspired by the other, or both are independently derivable from a common, more general formalism	Compression Test succeeds with low-to-moderate L_c , and a documented historical or formal lineage connects the two domains, even if no single equation unifies them.
Mechanism	A provable mathematical isomorphism exists between the two (D, T, R, C) structures — not merely similarity, but a demonstrable mapping with a citable proof	Compression Test succeeds with near-zero L_c , and an explicit mathematical derivation (not merely a verbal correspondence) connects the two structures. This is the gold standard and the only category RSP-2 treats as a strong scientific claim rather than a structural observation.

6.2 Retroactive Reclassification of Audit 3's Pairs

Applying this distinction to the ten expected-success pairs from Audit 3 changes their status. Genetic drift and Brownian motion are Mechanism-grade: the diffusion approximation to the Wright-Fisher model is a citable, provable mathematical result, not an analogy. The same is true of the SIR-to-mass-action correspondence and the Noether pairing between gauge invariance and charge conservation. By contrast, evolution and control theory share only Homology-grade status: both exhibit attractor-seeking dynamics under perturbation, and the resemblance is real and well-documented, but no single mathematical structure has been shown to underlie both — the case for treating them as the same formal object remains unproven.

This reclassification matters because RSP-1 made no distinction between these grades. Treating a Homology-grade resemblance (evolution and control theory) with the same confidence as a Mechanism-grade proof (drift and diffusion) is precisely the category error this audit was designed to catch. RSP-2 requires every reported translation to carry an explicit grade — Analogy, Homology, or Mechanism — and prohibits presenting Homology-grade or Analogy-grade results with Mechanism-grade confidence.

Audit 7 — Standard Model Stress Test

Task: Translate $SU(3) \times SU(2) \times U(1)$ into (D, T, R, C) form, then determine whether independent auditors would recover the same Rosetta Signature. Low inter-rater reliability would indicate the translation rules are underspecified.

7.1 Honest Limitation

This document cannot literally convene five independent auditors and measure agreement statistically. What it can do honestly is identify, from the inside, the specific points in the original RSP-1 Standard Model translation where a second auditor could reasonably have made a different choice — and propose the concrete mechanism by which inter-rater reliability would actually be measured and improved going forward.

7.2 Identified Points of Ambiguity

Decision Point	Choice Made in RSP-1	Reasonable Alternative a Second Auditor Might Choose
What is D for the U(1) layer?	“The phase relationship of a localized field”	“The field configuration itself,” treating phase as a property of D rather than D’s full content — a different but defensible scoping
What is T for each layer?	“Gauge transformation”, named generically	The specific group action (e.g., “local U(1) rotation” as opposed to the unqualified “gauge transformation”) — a more specific labeling an auditor trained in differential geometry might prefer
Is C separate from T?	Treated implicitly as part of T in RSP-1 (pre-Audit 1 revision)	Under RSP-2’s revised grammar, C must be coded separately as “the specific Lie group structure,” which RSP-1 did not do — this is not an alternative choice but a correction

7.3 Proposed Reliability Protocol

Rather than claim a reliability result that was not actually measured, RSP-2 commits to the following protocol as a prerequisite for any future claim that a Standard Model translation, or any other translation, is well-specified:

- A written codebook defining D_type, T_class, R_class, and C_class categories must exist and be distributed to auditors before coding begins.
- A minimum of three independently trained auditors code the same domain without consulting one another.
- Agreement is measured using Cohen’s kappa across each of the four categorical fields separately, with a pre-registered minimum acceptable threshold of $\kappa \geq 0.7$ (substantial

agreement) before a Rosetta Signature is treated as reliable enough for cross-domain comparison.

- Domains failing to reach $\kappa \geq 0.7$ are flagged as underspecified, and the codebook is revised — not the data — until reliability improves.

Audit 7 Verdict: this is formalized as an open empirical problem, not a resolved one. RSP-1's Standard Model translation, however suggestive, has not been validated for inter-rater reliability and should not be cited as more secure than that. The reliability protocol above is the actual deliverable of this audit.

Audit 8 — Compression Frontier

Question: What is the minimal irreducible grammar? Audit 1 concluded that four primitives — (D, T, R, C) — are necessary. This audit tests whether that conclusion is actually minimal by attempting successive reductions and requiring each reduction attempt to fail with a specific, named counterexample rather than an unexamined assertion.

8.1 Reduction Attempt: (D, T, R, C) → (D, T, R)

This is the reduction Audit 1 already tested directly. The attempt is to fold C into either T or D.

- Folding C into T: redefine T as “the already-constrained transformation space” rather than keeping C separate. This fails because it conflates two distinct things — what the system could do absent any constraint, and what limits exist independent of the dynamics. In evolution, a resource limit exists whether or not mutation and selection happen to be actively operating; it is a property of the environment's boundary conditions, not of the transformation rule. Folding C into T turns T from a clean operator into “operator plus boundary conditions,” which silently reintroduces exactly the kind of implicit importation Audit 1 was built to catch.
- Folding C into D: redefine D as “the constrained space of distinctions” rather than keeping C separate. This fails for a different domain: carrying capacity in evolution can plausibly be read as restricting which population states are reachable (a property of D's range), but a legal constraint on which contract amendments are valid is a restriction on the dynamics (T), not on the space of possible contracts (D) — contracts of any content can exist in principle; only certain transformations of them are legally valid. Since C plays the D-restricting role in one domain and the T-restricting role in another, it cannot be uniformly absorbed into either.

Result: the reduction from (D, T, R, C) to (D, T, R) fails in both directions, with a specific counterexample in each case. C is confirmed irreducible.

8.2 Reduction Attempt: (D, T, R) → (D, T)

This reduction asks whether R is itself dispensable — whether retention can be read off automatically once D and T are specified. This fails immediately: D and T alone describe a system that changes, but say nothing about what, if anything, survives the change. Two systems can share identical D and T while having entirely different R (the same population, with the same possible mutations, can be selected toward different fitness criteria depending on the environment). R carries independent information and cannot be derived from D and T alone.

8.3 Reduction Attempt: (D, T) → (D)

This reduction asks whether a bare distinction, without any transformation acting on it, is sufficient. This fails trivially: a distinction with no transformation acting on it is a static fact, not a dynamical system, and the entire content of RSP-2 concerns what happens to a distinction as it is acted upon.

Reducing to D alone collapses the framework into bare classification, which is not what RSP-2 claims to provide.

8.4 Conclusion

$$g_{min} = (D, T, R, C)$$

The minimal irreducible grammar contains exactly four primitives. Three reduction attempts were tested (removing C, removing R, removing T), and each failed with a specific counterexample rather than an unexamined assertion. This is the strongest available evidence, short of a fully general proof across all possible domains, that (D, T, R, C) is the correct minimal grammar rather than an arbitrarily chosen one.

RSP-2 Formal Specification

The following consolidates the outcomes of all eight audits into a single specification superseding RSP-1 in full.

F.1 Revised Grammar

$$G = (D, T, R, C)$$

Symbol	Name	Definition
D	Distinction Space	What is being differentiated, tracked, or held as a meaningful state in the domain.
T	Transformation Space	The operator, force, or process acting upon D, including its specific structure (not merely that a transformation exists, but which one).
R	Retention Criterion	What determines whether a given distinction survives the transformation — what must remain invariant, conserved, or stable.
C	Constraint	The restriction on the space of admissible states or transformations that exists independently of the transformation rule itself — resource limits, code structure, legal validity conditions, boundary conditions.

F.2 Explanatory-Loss Metric

$$L_c = 1 - (|P_X_{\text{recovered}}| / |P_X|), \text{ } P_X \text{ pre-registered before translation}$$

F.3 Equivalence and Confidence Metrics

$$d(R_{s1}, R_{s2}) = w_c \cdot H(R_{s1}, R_{s2}) + w_g \cdot |dim(T_1) - dim(T_2)| / \max(dim(T_1), dim(T_2))$$

$$S(R_{s1}, R_{s2}) = 1 - d(R_{s1}, R_{s2})$$

$$Confidence(X, Y) = S(R_{sX}, R_{sY}) \times (1 - L_c)$$

F.4 Falsification Criteria

- A proposed translation must be tested against the adversarial pair set (Audit 3) before being reported, including at least one pair expected to fail for a similar surface reason to the proposed pair (the Riemann/fitness-landscape test for false positives).
- A reported translation must state its grade — Analogy, Homology, or Mechanism — using the decision rules of Audit 6, and must not present a lower grade with higher-grade confidence.
- A reported translation must state L_c for an explicitly pre-registered prediction set, not a post hoc one.
- If a domain pair returns FALSE under the Compression Test, this must be reported as a finding, not discarded as a failed attempt — null results are data.

F.5 Predictive Framework

Any application of RSP-2 to a new domain pair is incomplete until it generates at least one prediction in the form specified in Audit 4: a domain possessing known D and T but a contested or unknown R, paired with a directional, falsifiable claim about what should be observed if a specific R is correct.

F.6 Inter-Rater Reliability Protocol

Any claimed Rosetta Signature for a domain must eventually be validated using the codebook-and-Cohen's-kappa procedure specified in Audit 7 ($\kappa \geq 0.7$ across each categorical field) before it is treated as reliable enough to support cross-domain comparison. Until validated, a signature is provisional.

F.7 Minimum Grammar Proof

The grammar (D, T, R, C) is minimal: removing any one of the four primitives produces a specific, demonstrated failure (Audit 8), and no fifth primitive among the tested candidates (S, I, A) survived the necessity test of Audit 1.

Verdict

The task set for this audit permitted exactly two outcomes: RSP must emerge either as a falsifiable scientific translation framework, or as a useful heuristic that cannot support scientific claims. No third option was permitted.

What the Audit Found

- RSP-1's three-primitive grammar was insufficient and required a fourth primitive, C, proven necessary rather than assumed (Audits 1 and 8).
- RSP-1's central evaluative criterion — translation without explanatory loss — was unmeasurable as stated and has been replaced with a computable metric that explicitly acknowledges its own scope limits, including a demonstrated case (QED's quantitative content) where loss is necessarily high (Audit 2).
- RSP-1 had no mechanism for returning FALSE. RSP-2 does, and the adversarial pair set in Audit 3 demonstrates the mechanism working correctly on ten pairs designed to fail, each failing at a specifically named stage.
- RSP-1 could not distinguish a provable mathematical correspondence from a suggestive verbal analogy. RSP-2's three-way grading system (Analogy / Homology / Mechanism) repairs this and forces every reported translation to carry an explicit confidence grade.
- RSP-1's claim to have translated the Standard Model was never tested for whether independent investigators would reconstruct the same translation. This remains an open empirical problem; RSP-2 commits to a specific protocol for resolving it rather than asserting a result that was not measured.

Final Determination

RSP-2 emerges from this audit as a falsifiable scientific translation framework, conditional on three commitments that were not present in RSP-1: every reported translation must carry an explicit grade and a computed loss score; every application to a new domain pair must include at least one directional, falsifiable prediction; and every claimed Rosetta Signature remains provisional until validated under the inter-rater reliability protocol. Where this audit could not produce a closed result — full automation of the signature metric, and empirical inter-rater validation of the Standard Model translation — the gap is recorded as an open problem with a specified resolution path, not concealed behind confident language. A framework that can state its own unresolved weaknesses this precisely is doing something other than producing an unfalsifiable analogy machine, which was the central risk this audit was designed to test for.